

Traitle Y

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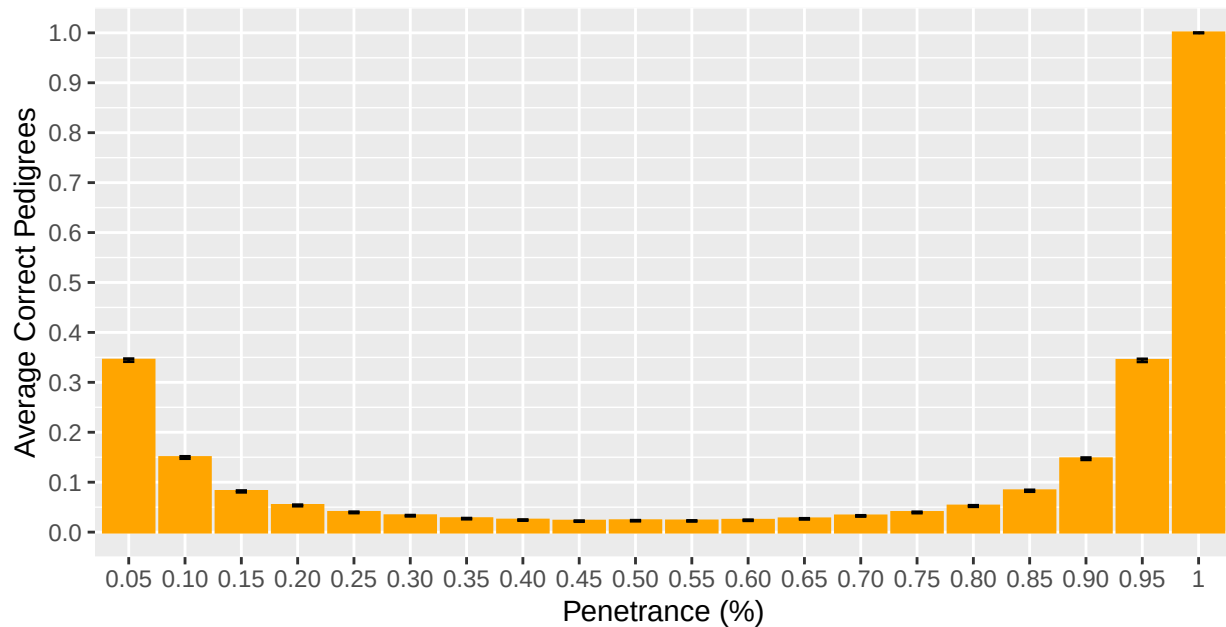
2026-03-18

```
library("ggplot2")
library("plotrix")
library("stringr")
all.data = read.csv("results.csv")
diseaseType = all.data$DiseaseType
penetrance = all.data$Penetrance
correct = as.integer(all.data$Correct=="true")
cnt = 0
M.corrected = c()
M.sd = c()
start = 1
end = 100000
for(i in 1:20){
  M.corrected = c(M.corrected, mean(correct[start:end]))
  M.sd = c(M.sd, 1.96*std.error(correct[start:end]))
  start = start + 100000
  end = end + 100000
}

dfM <- data.frame(Penetrance = c("1", "0.95", "0.90", "0.85", "0.80", "0.75", "0.70", "0.65", "0.60", "0.55", "0.50", "0.45", "0.40", "0.35", "0.30", "0.25", "0.20", "0.15", "0.10", "0.05", "0.00"))
ggplot(dfM, aes(x = Penetrance, y = AverageCorrect)) +
  geom_bar(stat = "identity", fill = "orange", color = "orange") +
  labs(
    x = str_wrap("Penetrance (%)"),
    y = str_wrap("Average Correct Pedigrees"),
    title = str_wrap("Classification Accuracy for Mitochondrial Pedigrees with different levels of Penetrance"),
    subtitle = str_wrap("Traitle Y"),
    caption = str_wrap("The parental generation consists of 2 individuals. In the F1 generation, each parent is affected by the disease with a probability of 0.5")
  ) +
  theme(
    plot.title = element_text(hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5),
    plot.caption = element_text(hjust = 0.5)
  ) +
  geom_errorbar(aes(ymin = AverageCorrect - sd, ymax = AverageCorrect + sd), width = 0.2) +
  scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, by = 0.1))
```

Classification Accuracy for Mitochondrial Pedigrees with different levels of Penetrance

Traitle Y



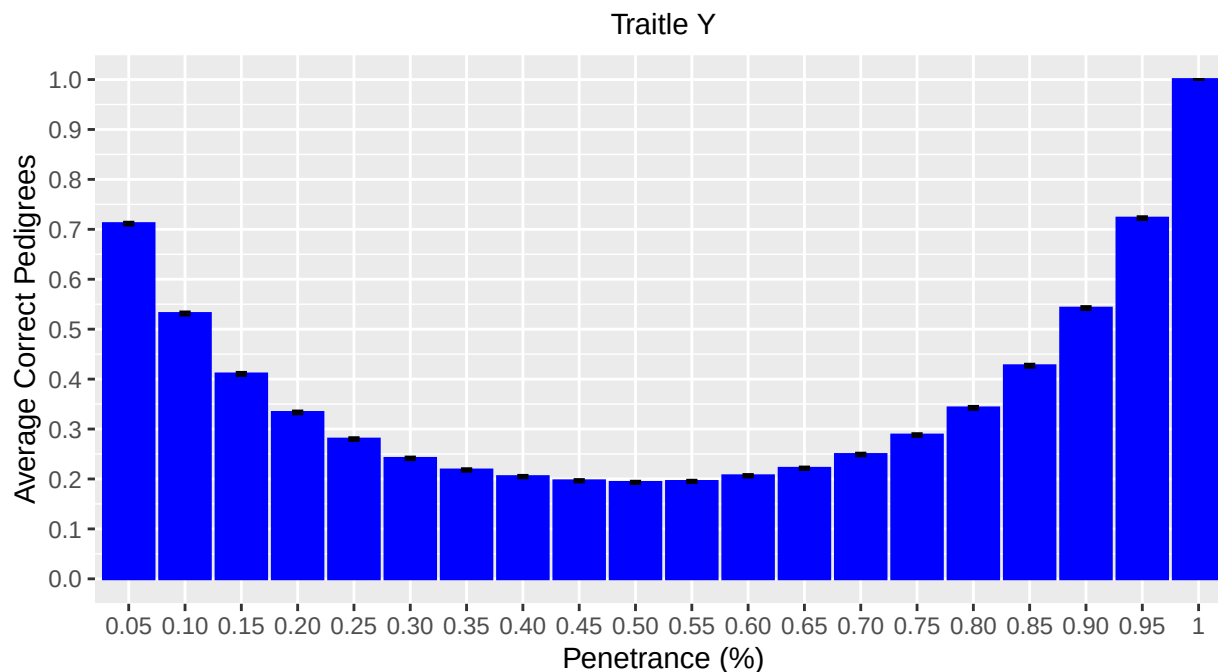
The parental generation consists of 2 individuals. In the F1 generation, each parent has 3–4 children, and in the F2 generation, each F1 individual has 5–7 children.

```
XLD.corrected = c()
XLD.sd = c()
for(i in 1:20){
  XLD.corrected = c(XLD.corrected, mean(correct[start:end]))
  XLD.sd = c(XLD.sd, 1.96*std.error(correct[start:end]))
  start = start + 100000
  end = end + 100000
}

dfXLD <- data.frame(Penetrance = c("1", "0.95", "0.90", "0.85", "0.80", "0.75", "0.70", "0.65", "0.60",
ggplot(dfXLD, aes(x = Penetrance, y = AverageCorrect)) +
  geom_bar(stat = "identity", fill = "blue", color = "blue") +
  labs(
    x = str_wrap("Penetrance (%)" ),
    y = str_wrap("Average Correct Pedigrees"),
    title = str_wrap("Classification Accuracy for X-Linked Dominant Pedigrees with different levels of I",
    subtitle = str_wrap("Traitle Y"),
    caption = str_wrap("The parental generation consists of 2 individuals. In the F1 generation, each p
  ) +
  theme(
    plot.title = element_text(hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5),
    plot.caption = element_text(hjust = 0.5)
  ) +
  geom_errorbar(aes(ymin = AverageCorrect - sd, ymax = AverageCorrect + sd), width = 0.2) +
```

```
scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, by = 0.1))
```

Classification Accuracy for X-Linked Dominant Pedigrees with different levels Penetrance

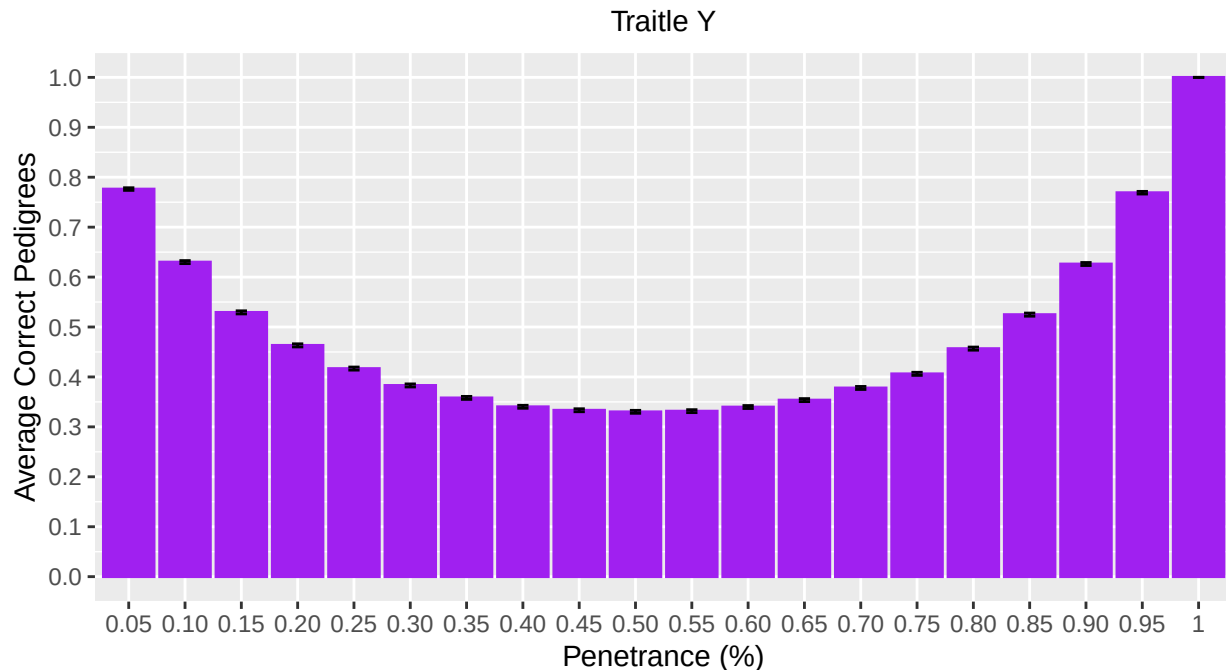


The parental generation consists of 2 individuals. In the F1 generation, each parent has 3–4 children, and in the F2 generation, each F1 individual has 5–7 children.

```
XLR.corrected = c()
XLR.sd = c()
for(i in 1:20){
  XLR.corrected = c(XLR.corrected, mean(correct[start:end]))
  XLR.sd = c(XLR.sd, 1.96*std.error(correct[start:end]))
  start = start + 100000
  end = end + 100000
}
dfXLR <- data.frame(Penetrance = c("1", "0.95", "0.90", "0.85", "0.80", "0.75", "0.70", "0.65", "0.60",
ggplot(dfXLR, aes(x = Penetrance, y = AverageCorrect)) +
  geom_bar(stat = "identity", fill = "purple" , color = "purple") +
  labs(
    x = str_wrap("Penetrance (%)"),
    y = str_wrap("Average Correct Pedigrees"),
    title = str_wrap("Classification Accuracy for X-Linked Recessive Pedigrees with different levels of",
    subtitle = str_wrap("Traitle Y"),
    caption = str_wrap("The parental generation consists of 2 individuals. In the F1 generation, each p
  ) +
  theme(
    plot.title = element_text(hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5),
    plot.caption = element_text(hjust = 0.5)
  ) +
```

```
geom_errorbar(aes(ymin = AverageCorrect - sd, ymax = AverageCorrect + sd), width = 0.2) +
scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, by = 0.1))
```

Classification Accuracy for X-Linked Recessive Pedigrees with different level: of Penetrance

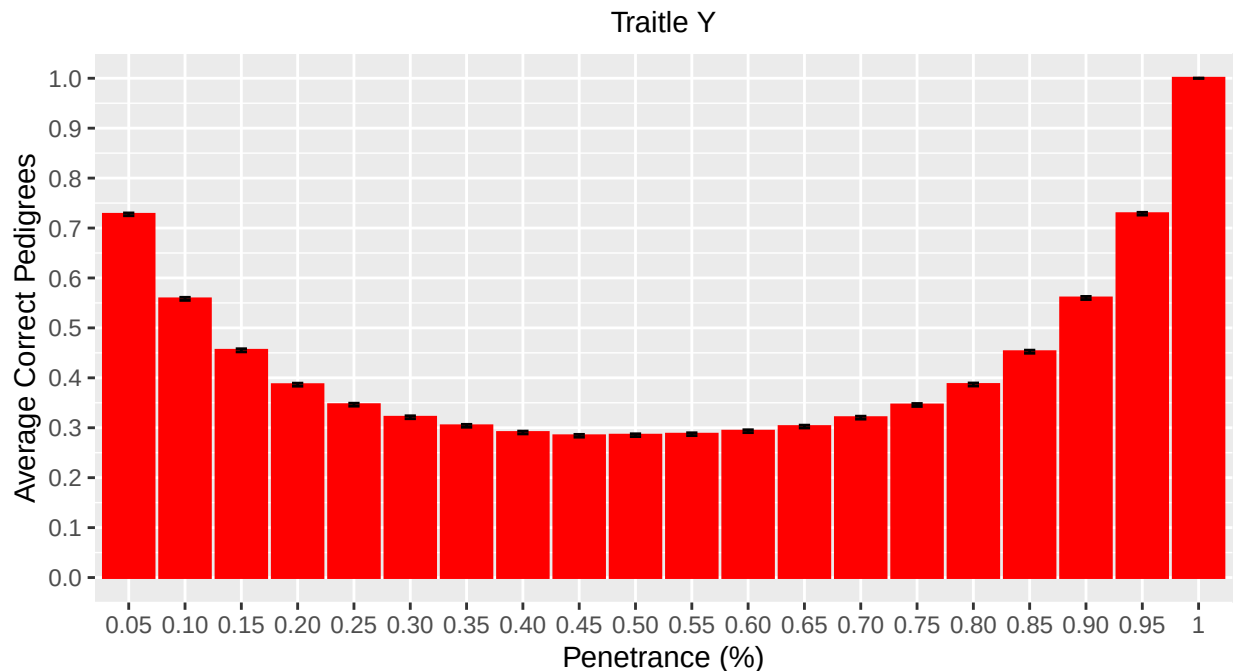


The parental generation consists of 2 individuals. In the F1 generation, each parent has 3–4 children, and in the F2 generation, each F1 individual has 5–7 children.

```
YL.corrected = c()
YL.sd = c()
for(i in 1:20){
  YL.corrected = c(YL.corrected, mean(correct[start:end]))
  YL.sd = c(YL.sd, 1.96*std.error(correct[start:end]))
  start = start + 100000
  end = end + 100000
}
dfYL <- data.frame(Penetrance = c("1", "0.95", "0.90", "0.85", "0.80", "0.75", "0.70", "0.65", "0.60", "0.55", "0.50", "0.45", "0.40", "0.35", "0.30", "0.25", "0.20", "0.15", "0.10", "0.05"))
ggplot(dfYL, aes(x = Penetrance, y = AverageCorrect)) +
  geom_bar(stat = "identity", fill = "red", color = "red") +
  labs(
    x = str_wrap("Penetrance (%)" ),
    y = str_wrap("Average Correct Pedigrees"),
    title = str_wrap("Classification Accuracy for Y-Linked Pedigrees with different levels of Penetrance"),
    subtitle = str_wrap("Traitle Y"),
    caption = str_wrap("The parental generation consists of 2 individuals. In the F1 generation, each parent has 3–4 children, and in the F2 generation, each F1 individual has 5–7 children.")
  ) +
  theme(
    plot.title = element_text(hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5),
    plot.caption = element_text(hjust = 0.5)
```

```
)+
geom_errorbar(aes(ymin = AverageCorrect - sd, ymax = AverageCorrect + sd), width = 0.2)+
scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, by = 0.1))
```

Classification Accuracy for Y-Linked Pedigrees with different levels of Penetrance



The parental generation consists of 2 individuals. In the F1 generation, each parent has 3–4 children, and in the F2 generation, each F1 individual has 5–7 children.

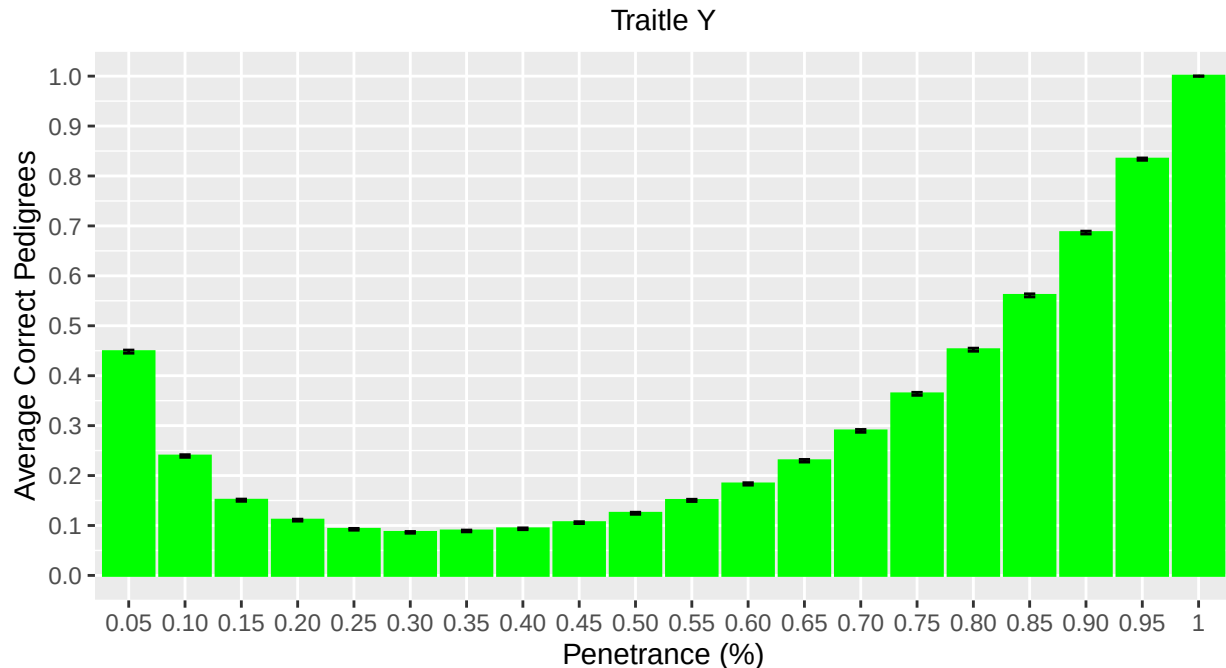
```
AD.corrected = c()
AD.sd = c()
for(i in 1:20){
  AD.corrected = c(AD.corrected, mean(correct[start:end]))
  AD.sd = c(AD.sd, 1.96*std.error(correct[start:end]))
  start = start + 100000
  end = end + 100000
}
dfAD <- data.frame(Penetrance = c("1", "0.95", "0.90", "0.85", "0.80", "0.75", "0.70", "0.65", "0.60", "0.55", "0.50", "0.45", "0.40", "0.35", "0.30", "0.25", "0.20", "0.15", "0.10", "0.05"))
ggplot(dfAD, aes(x = Penetrance, y = AverageCorrect)) +
  geom_bar(stat = "identity", fill = "green", color = "green") +
  labs(
    x = str_wrap("Penetrance (%)"),
    y = str_wrap("Average Correct Pedigrees"),
    title = str_wrap("Classification Accuracy for Autosomal Dominant Pedigrees with different levels of Penetrance"),
    subtitle = str_wrap("Traitle Y"),
    caption = str_wrap("The parental generation consists of 2 individuals. In the F1 generation, each parent has 3–4 children, and in the F2 generation, each F1 individual has 5–7 children.")
  ) +
  theme(
    plot.title = element_text(hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5),
    plot.caption = element_text(hjust = 0.5)
```

```

plot.caption = element_text(hjust = 0.5)
)+
geom_errorbar(aes(ymin = AverageCorrect - sd, ymax = AverageCorrect + sd), width = 0.2)+
scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, by = 0.1))

```

Classification Accuracy for Autosomal Dominant Pedigrees with different level of Penetrance



The parental generation consists of 2 individuals. In the F1 generation, each parent has 3–4 children, and in the F2 generation, each F1 individual has 5–7 children.

```

AR.corrected = c()
AR.sd = c()
for(i in 1:20){
  AR.corrected = c(AR.corrected, mean(correct[start:end]))
  AR.sd = c(AR.sd, 1.96*std.error(correct[start:end]))
  start = start + 100000
  end = end + 100000
}
dfAR <- data.frame(Penetrance = c("1", "0.95", "0.90", "0.85", "0.80", "0.75", "0.70", "0.65", "0.60", "0.55", "0.50", "0.45", "0.40", "0.35", "0.30", "0.25", "0.20", "0.15", "0.10", "0.05"))
ggplot(dfAR, aes(x = Penetrance, y = AverageCorrect)) +
  geom_bar(stat = "identity", fill = "pink", color = "pink") +
  labs(
    x = str_wrap("Penetrance (%)"),
    y = str_wrap("Average Correct Pedigrees"),
    title = str_wrap("Classification Accuracy for Autosomal Recessive Pedigrees with different levels of Penetrance"),
    subtitle = str_wrap("Traitle Y"),
    caption = str_wrap("The parental generation consists of 2 individuals. In the F1 generation, each parent has 3–4 children, and in the F2 generation, each F1 individual has 5–7 children.")
  ) +
  theme(
    plot.title = element_text(hjust = 0.5),

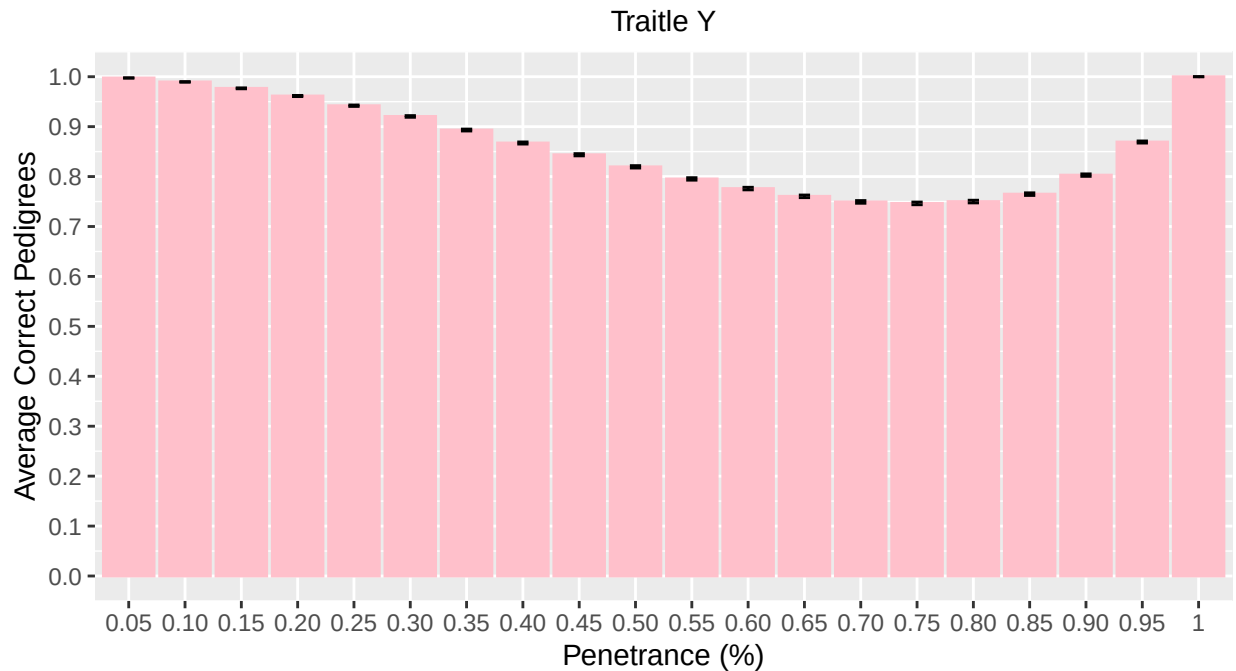
```

```

plot.subtitle = element_text(hjust = 0.5),
plot.caption = element_text(hjust = 0.5)
)+
geom_errorbar(aes(ymin = AverageCorrect - sd, ymax = AverageCorrect + sd), width = 0.2)+
scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, by = 0.1))

```

Classification Accuracy for Autosomal Recessive Pedigrees with different level of Penetrance



The parental generation consists of 2 individuals. In the F1 generation, each parent has 3–4 children, and in the F2 generation, each F1 individual has 5–7 children.

```

all.data$Correct = as.numeric(as.logical(all.data$Correct))

logistic.all <- glm(Correct ~ DiseaseType * (Penetrance + I(Penetrance^2)), data=all.data, family="binomial")
summary(logistic.all)

```

```

##
## Call:
## glm(formula = Correct ~ DiseaseType * (Penetrance + I(Penetrance^2)),
##     family = "binomial", data = all.data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.088392   0.005549  -15.93  <2e-16 ***
## DiseaseTypeAR    6.095788   0.016425  371.12  <2e-16 ***
## DiseaseTypeM     0.514075   0.008618   59.65  <2e-16 ***
## DiseaseTypeXLD    1.454441   0.007524  193.31  <2e-16 ***
## DiseaseTypeXLR    1.647567   0.007561  217.90  <2e-16 ***
## DiseaseTypeYL     1.366843   0.007469  183.00  <2e-16 ***
## Penetrance    -10.865344   0.027136 -400.40  <2e-16 ***

```

```
## I(Penetrance^2)          13.546087    0.026709   507.17   <2e-16 ***
## DiseaseTypeAR:Penetrance -4.065973    0.059243   -68.63   <2e-16 ***
## DiseaseTypeM:Penetrance -10.698594    0.051542  -207.57   <2e-16 ***
## DiseaseTypeXLD:Penetrance -1.502488    0.036304   -41.39   <2e-16 ***
## DiseaseTypeXLR:Penetrance  0.867060    0.035663    24.31   <2e-16 ***
## DiseaseTypeYL:Penetrance  0.849557    0.035512    23.92   <2e-16 ***
## DiseaseTypeAR:I(Penetrance^2) -2.250326    0.050125   -44.89   <2e-16 ***
## DiseaseTypeM:I(Penetrance^2)  8.667435    0.049696   174.41   <2e-16 ***
## DiseaseTypeXLD:I(Penetrance^2) -0.859715    0.035210   -24.42   <2e-16 ***
## DiseaseTypeXLR:I(Penetrance^2) -3.352595    0.034546   -97.05   <2e-16 ***
## DiseaseTypeYL:I(Penetrance^2) -3.258908    0.034396   -94.75   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 16420268  on 11999999  degrees of freedom
## Residual deviance: 11536679  on 11999982  degrees of freedom
## AIC: 11536715
##
## Number of Fisher Scoring iterations: 6
```

```
M.log <- all.data %>%
  dplyr::filter(DiseaseType=="M")
logistic.M <- glm(Correct ~ Penetrance + I(Penetrance^2), data=M.log, family="binomial")
summary(logistic.M)
```

```
##
## Call:
## glm(formula = Correct ~ Penetrance + I(Penetrance^2), family = "binomial",
## data = M.log)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    0.425683   0.006594   64.56   <2e-16 ***
## Penetrance    -21.563938   0.043820 -492.10   <2e-16 ***
## I(Penetrance^2) 22.213522   0.041908  530.05   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 1533331  on 1999999  degrees of freedom
## Residual deviance:  997690  on 1999997  degrees of freedom
## AIC: 997696
##
## Number of Fisher Scoring iterations: 6
```

```
XLD.log <- all.data %>%
  dplyr::filter(DiseaseType=="XLD")
logistic.XLD <- glm(Correct ~ Penetrance + I(Penetrance^2), data=XLD.log, family="binomial")
summary(logistic.XLD)
```

```
##
## Call:
## glm(formula = Correct ~ Penetrance + I(Penetrance^2), family = "binomial",
##      data = XLD.log)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    1.366049   0.005081   268.8   <2e-16 ***
## Penetrance     -12.367832   0.024116  -512.8   <2e-16 ***
## I(Penetrance^2)  12.686372   0.022942   553.0   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 2647660  on 1999999  degrees of freedom
## Residual deviance: 2259354  on 1999997  degrees of freedom
## AIC: 2259360
##
## Number of Fisher Scoring iterations: 4
```

```
XLR.log <- all.data %>%
  dplyr::filter(DiseaseType=="XLR")
logistic.XLR <- glm(Correct ~ Penetrance + I(Penetrance^2), data=XLR.log, family="binomial")
summary(logistic.XLR)
```

```
##
## Call:
## glm(formula = Correct ~ Penetrance + I(Penetrance^2), family = "binomial",
##      data = XLR.log)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    1.559175   0.005136   303.6   <2e-16 ***
## Penetrance     -9.998284   0.023140  -432.1   <2e-16 ***
## I(Penetrance^2) 10.193492   0.021911   465.2   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 2771297  on 1999999  degrees of freedom
## Residual deviance: 2509737  on 1999997  degrees of freedom
## AIC: 2509743
##
## Number of Fisher Scoring iterations: 4
```

```
YL.log <- all.data %>%
  dplyr::filter(DiseaseType=="YL")
logistic.YL <- glm(Correct ~ Penetrance + I(Penetrance^2), data=YL.log, family="binomial")
summary(logistic.YL)
```

```
##
```

```
## Call:
## glm(formula = Correct ~ Penetrance + I(Penetrance^2), family = "binomial",
##      data = YL.log)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    1.27845    0.00500   255.7  <2e-16 ***
## Penetrance    -10.01579    0.02291  -437.2  <2e-16 ***
## I(Penetrance^2) 10.28718    0.02167   474.6  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 2735126  on 1999999  degrees of freedom
## Residual deviance: 2461385  on 1999997  degrees of freedom
## AIC: 2461391
##
## Number of Fisher Scoring iterations: 4
```

```
AD.log <- all.data %>%
  dplyr::filter(DiseaseType=="AD")
logistic.AD <- glm(Correct ~ Penetrance + I(Penetrance^2), data=AD.log, family="binomial")
summary(logistic.AD)
```

```
##
## Call:
## glm(formula = Correct ~ Penetrance + I(Penetrance^2), family = "binomial",
##      data = AD.log)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -0.088392    0.005549  -15.93  <2e-16 ***
## Penetrance    -10.865344    0.027136 -400.40  <2e-16 ***
## I(Penetrance^2) 13.546087    0.026709  507.18  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 2490530  on 1999999  degrees of freedom
## Residual deviance: 1845229  on 1999997  degrees of freedom
## AIC: 1845235
##
## Number of Fisher Scoring iterations: 4
```

```
AR.log <- all.data %>%
  dplyr::filter(DiseaseType=="AR")
logistic.AR <- glm(Correct ~ Penetrance + I(Penetrance^2), data=AR.log, family="binomial")
summary(logistic.AR)
```

```
##
## Call:
```

```
## glm(formula = Correct ~ Penetrance + I(Penetrance^2), family = "binomial",
##     data = AR.log)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)      6.00740    0.01546   388.6  <2e-16 ***
## Penetrance     -14.93132    0.05266  -283.5  <2e-16 ***
## I(Penetrance^2)  11.29576    0.04242   266.3  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1610646  on 1999999  degrees of freedom
## Residual deviance: 1463284  on 1999997  degrees of freedom
## AIC: 1463290
##
## Number of Fisher Scoring iterations: 6
```